### Receptor-Mediated Endocytosis

Receptor-mediated endocytosis is a targeted process, unlike endocytosis in general, where receptor molecules are employed on the outer surface of the membrane to bind to a substance that need to be transported into the cell. The receptor exhibits specific binding affinity towards the substance that accomplishes the process. A protein coated vesicle called clathrin coated vesicle is attached to the cytosolic side of the membrane and this would pinch off from the membrane once the substance is transported into the coated vesicle.

Membrane uses a process called pinocytosis, a varied process of endocytosis. Pinocytosis uses smaller vesicles and occur in molecules (including liquid) whenever required by the cell from the extracellular fluid. Another varied process of pinocytosis called Potocytosis, uses a coating protein called caveolin, whose function is same as that of clathrin. In addition to caveolin, membranes contain cavities which form vacuoles and have receptors as well as lipid rafts.

In liver cell, receptor-mediated endocytosis is involved in the uptake of a type of cholesterol, called low-density lipoprotein or LDL bad cholesterol. If the receptor-mediated endocytosis is unsuccessful, bad cholesterol will not be removed and gets accumulated. This may result in familial hypercholesterolemia

## 1.4.VISCOSITY AND SURFACE TENSION

One of the most amazing features of biological membranes is that both the lipid and protein moieties are persistently in motion. This is the property of any molecule of a viscous liquid. The membrane acts as a two-dimensional liquid in which the protein constituents stroll like boats.

The lipid bilayer behaves as a two-dimensional liquid in the absence of covalent bonds between the lipids, whose resistance against shear deformations is characterized by the surface shear viscosity. Any relative motion between the two leaflets of a bilayer is opposed by a frictional force. Living cells actively control these forces by varying the mixture of lipids and sterols present in their membranes

Ouabain, is a plant derived toxic substance and in eastern Africa it was used as an arrow poison warfare and in hunting during ancient times. Ouabain is a cardiac glycoside and at low doses can be useful for treating arrhythmia and hypotension. It can inhibit the ATPase activity.

### 1.4.1.Biological importance of viscosity and surface tension

1. Blood viscosity is useful in streamlining the blood flow. Blood plasma has a normal viscosity of 15 – 20 mpoises. Alterations in the viscosity are an indication of diseased condition. Viscosity increases during macroglobulinemia, retinal hemorrhages and congestive heart failure.



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